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# RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

III Semester B. E. Supplementary/Makeup Examinations October-2024

Common to EC/ EE/ EI/ET

## LINEAR ALGEBRA, FOURIER TRANSFORMS AND STATISTICS

*Time: 03 Hours* *Maximum Marks: 100*

*Instructions to candidates:*

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of scientific calculator and hand book for mathematics is permitted.

### PART-A

| | | | M | CO | BT |
|---|---|---|---|---|---|
| 1 | 1.1 | The value of $k$ such that the vectors $(1,2,1),(1,1,-1),(4,k,-2)$ are linearly dependent is _____. | 01 | 2 | 2 |
| | 1.2 | If $S_1$ and $S_2$ are two subspaces of vector space $V$, then $S_1 \cap S_2$ is _____ of vector space $V$. | 01 | 1 | 1 |
| | 1.3 | Fourier series of the function $f(x) = \begin{cases} x+2, -2<x<0 \\ \sin x,\ 0<x<2 \end{cases}$ at the point of discontinuity converges to _____. | 01 | 2 | 1 |
| | 1.4 | In Fourier series expansion of $f(x) = x^2$ over the interval $(0,2)$, $a_0$ is _____. | 01 | 1 | 1 |
| | 1.5 | Fourier cosine transform of $f(x) = \begin{cases} 1, & 0<x<\pi \\ 0, & otherwise \end{cases}$ is _____. | 02 | 1 | 1 |
| | 1.6 | The lines of regressions coincide if correlation coefficient $r =$_____. | 01 | 1 | 2 |
| | 1.7 | If $\beta_2 = 3$, then distribution is _____. | 01 | 2 | 2 |
| | 1.8 | Find the image of the vector $(-3,4)$ when it is rotated through an angle of $30^0$. | 02 | 1 | 1 |
| | 1.9 | In QR factorization of the matrix $A = \begin{bmatrix} 5 & 9 \\ 1 & 7 \end{bmatrix}$, if $Q = \begin{bmatrix} \frac{5}{\sqrt{26}} & -\frac{1}{\sqrt{26}} \\ \frac{1}{\sqrt{26}} & \frac{5}{\sqrt{26}} \end{bmatrix}$, then $R$ is _____. | 02 | 2 | 3 |
| | 1.10 | Given matrix $A = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$, then singular value of the matrix $A$ is _____. | 02 | 1 | 2 |
| | 1.11 | Express the graph of the function $f(x) = \lvert x \rvert$ in $(-\pi,\pi)$ periodically. | 02 | 2 | 2 |
| | 1.12 | If Fourier transform of $f(x)$ is $\frac{1}{s^2+1}$, then the Fourier transform of $\{f(2x)\}\cos 3x$ is _____. | 02 | 2 | 2 |
| | 1.13 | In a partially destroyed laboratory record, only the lines of regression of $y$ on $x$ and $x$ on $y$ are available as $4x - 5y + 33 = 0$, $20x - 9y = 107$ respectively. Calculate $\bar{x}$ and $\bar{y}$. | 02 | 2 | 3 |

### PART-B

| | | | | | |
|---|---|---|---|---|---|
| 2 | a | Let $S = \{2x^2 - 5, 1 + x\}$, show that the polynomial $4x^2 + 3x - 7$ lies in $[S]$. | 04 | 2 | 2 |
| | b | Find the linear transformation $T: R^2 \to R^3$ such that $T(1,1) = (0,1,2)$ and $T(-1,1) = (2,1,0)$. | 04 | 3 | 3 |

| | | | | | |
|---|---|---|---|---|---|
| | c | Identify the bases and dimensions of the row space, column space and null space of the matrix $A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & -5 & 1 \\ 1 & -2 & 2 \\ 0 & 1 & -1 \end{bmatrix}$ | 08 | 3 | 2 |
| 3 | a | Diagonalize the matrix $B = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ and hence compute $B^8$. | 08 | 3 | 3 |
| | b | Determine the least square solution for the inconsistent system $AX = B$ where $A = \begin{bmatrix} 1 & -6 \\ 1 & -2 \\ 1 & 1 \\ 1 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} -1 \\ 2 \\ 1 \\ 6 \end{bmatrix}$ | 08 | 2 | 2 |
| | | **OR** | | | |
| 4 | a | Find the singular value decomposition of the matrix $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$ | 08 | 3 | 3 |
| | b | Applying Gram-Schmidt process, obtain the orthonormal basis for the column space the matrix $C = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$. | 08 | 2 | 2 |
| 5 | a | Express the following function $f(x) = \begin{cases} 1 + \left(\frac{2x}{\pi}\right), & -\pi < x < 0 \\ 1 - \left(\frac{2x}{\pi}\right), & 0 \le x < \pi \end{cases}$ as Fourier series. Deduce that $\frac{\pi^2}{8} = \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2}$. | 08 | 2 | 3 |
| | b | Expand the function $f(x) = e^x$ over the interval (0,1) in half range Fourier cosine series and in half range Fourier sine series. | 08 | 3 | 4 |
| | | **OR** | | | |
| 6 | a | Obtain the complex Fourier series of the function $f(x) = \begin{cases} 0, & 0 < x < l \\ a, & l < x < 2l \end{cases}$ over the interval $(0,2l)$. Deduce the corresponding trigonometric Fourier series. | 08 | 2 | 3 |
| | b | The following table gives the values of a periodic current $I$ (in approximate units) measured at 12 equidistant values of $\theta$ covering a period of $2\pi$. | | | |

| $\theta$ | 0 | $\frac{\pi}{6}$ | $\frac{\pi}{3}$ | $\frac{\pi}{2}$ | $\frac{2\pi}{3}$ | $\frac{5\pi}{6}$ | $\pi$ | $\frac{7\pi}{6}$ | $\frac{4\pi}{3}$ | $\frac{3\pi}{2}$ | $\frac{5\pi}{3}$ | $\frac{11\pi}{6}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $I$: | 0 | 2.3 | 5.5 | 8.9 | 10.8 | 11.4 | 9.9 | 4.8 | 0 | 0 | 0 | 0 |

| | | | | | |
|---|---|---|---|---|---|
| | | Express $I$ in the Fourier series as far as the first harmonic. | 08 | 3 | 4 |
| 7 | a | Obtain the complex Fourier transform of the function $f(x) = \begin{cases} 1, & \lvert x \rvert \le a \\ 0, & \lvert x \rvert > a \end{cases}$ where $a$ is a positive constant. Hence evaluate $\int_0^{\infty} \frac{\sin x}{x} dx$. | 08 | 4 | 4 |
| | b | Find the Fourier sine and cosine transform of the following function: $f(x) = \begin{cases} x, & 0 < x < 2 \\ 0, & elsewhere \end{cases}$ | 08 | 2 | 3 |
| | | **OR** | | | |

| | | | | | |
|---|---|---|---|---|---|
| 8 | a | Determine the Fourier transform of the function $e^{-a^2x^2}, a > 0$. Hence, deduce that $e^{-\frac{x^2}{2}}$ is self-reciprocal in respect of Fourier transform. | 08 | 4 | 4 |
| | b | Find the Fourier sine transform of $e^{-ax}$, and then evaluate $\int_0^\infty \frac{t^2}{(t^2+4)(t^2+9)} dt$ using Parseval identity. | 08 | 2 | 3 |
| 9 | a | In a certain distribution, the first 4 moment about the point 4 are $-1.5, 17, -30$ and 108. Cal culate the moments about the mean. Also evaluate $\beta_1, \beta_2$ and comment upon the skewness and kurtosis of the distribution. | 08 | 2 | 2 |
| | b | The following table shows the ages $x$ and the systolic pressure of 6 persons.<br>Age ($x$): 56, 42, 72, 36, 63, 47<br>Blood pressure ($y$): 147, 125, 160, 118, 149, 128<br>Compute the coefficient of correlation between $x$ and $y$. Estimate the blood pressure of a person whose age is 45*years*. | 08 | 3 | 4 |
| | | **OR** | | | |
| 10 | a | Calculate rank correlation coefficient from the following data:<br>$x$: 81, 78, 73, 73, 69, 68, 62, 58<br>$y$: 10, 12, 18, 18, 18, 22, 20, 24 | 08 | 2 | 2 |
| | b | Estimate $y$ for given $x_1 = 9$ and $x_2 = 8$ by fitting a regression plane $y = a_0 + a_1x_1 + a_2x_2$ to the following data:<br>$y$: 90, 72, 54, 42, 30, 12<br>$x_1$: 3, 5, 6, 8, 12, 14<br>$x_2$: 16, 10, 7, 4, 3, 2 | 08 | 3 | 4 |